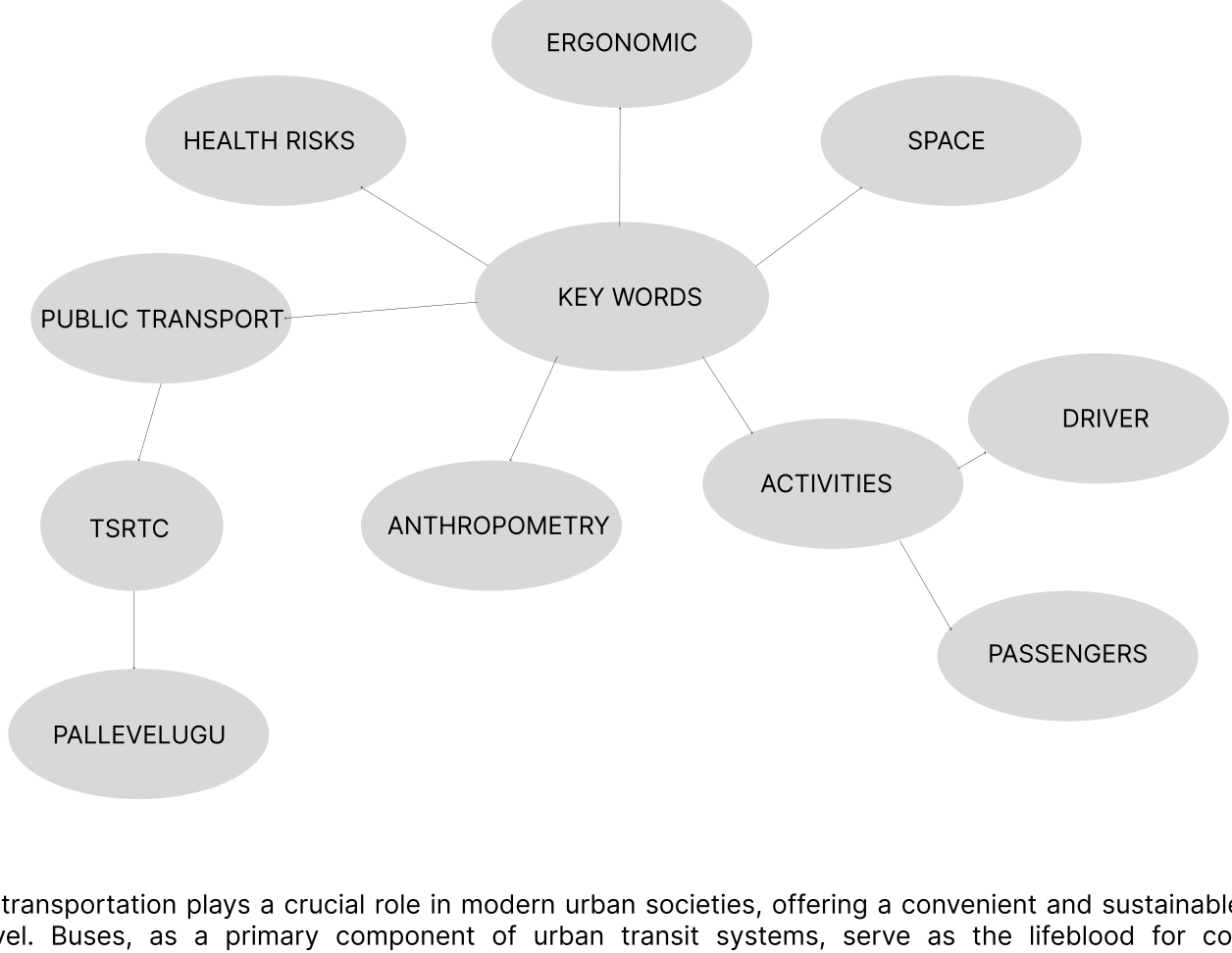


A comprehensive analysis of spatial utilization in a state transport bus through activity mapping of drivers and passengers .



Public transportation plays a crucial role in modern urban societies, offering a convenient and sustainable mode of travel. Buses, as a primary component of urban transit systems, serve as the lifeblood for countless commuters worldwide. However, the overall health and comfort of passengers and drivers are significantly influenced by the spatial design of the bus, the activities taking place within it, and the associated health risks stemming from ergonomic factors. **This research paper aims to investigate the various spaces within a bus, the activities of passengers and drivers, and explore potential interventions for improving bus ergonomics. Moreover, it seeks to illuminate the health risks associated with the spatial utilization and design of buses, thereby paving the way for informed, practical recommendations.**

Efficient spatial utilization and ergonomic considerations are not only essential for the comfort of passengers but also for the well-being of bus drivers. Passengers often spend extended periods on buses, making it imperative to analyze how they interact with the available spaces, seating arrangements, and amenities. Additionally, bus drivers face unique ergonomic challenges, as they spend long hours navigating through urban traffic, which can take a toll on their health. By studying the spatial dynamics within buses and the activities of both passengers and drivers, we can identify areas for improvement and develop evidence-based suggestions to enhance the overall bus experience.

Understanding the intricacies of bus ergonomics and health risks requires a multifaceted approach, encompassing aspects such as seat design, aisle space, safety measures, and driver well-being. This research paper will systematically explore these elements, drawing on a combination of empirical data, expert opinions, and relevant literature. Through a comprehensive analysis of the bus environment and the activities occurring within it, we aim to provide valuable insights into the spatial optimization of buses, thereby contributing to the enhancement of passenger and driver well-being.

In the following sections, we will delve into the specific components of bus ergonomics, the associated health risks, and potential interventions aimed at making bus travel safer, more comfortable, and sustainable. Ultimately, this research paper aspires to foster discussions and innovations that will benefit both the passengers who rely on public transportation and the dedicated drivers who ensure its operation.

METHODS

OBSERVATION

The activities of the Telangana bus drivers and the passengers encompassing actions both inside and outside the bus, were observed for two days. This included ingress and egress, sitting, standing, and movement around the bus, by both drivers and passengers; how the driver accessed the switchboard, side mirror, horn, wiper, ignition, and dashboard controls and so on.The analysis aimed to identify potential areas for improvement.

DATA COLLECTION

Measurement

The Telangana bus's interior and exterior were accurately measured using a measuring tape. Subsequently, the anthropometric measures of both drivers and passengers were recorded. The correlation between these two sets of measurements was utilized to pinpoint areas posing potential ergonomic issues.

Anthropometric measure checklist (Driver and Passenger)

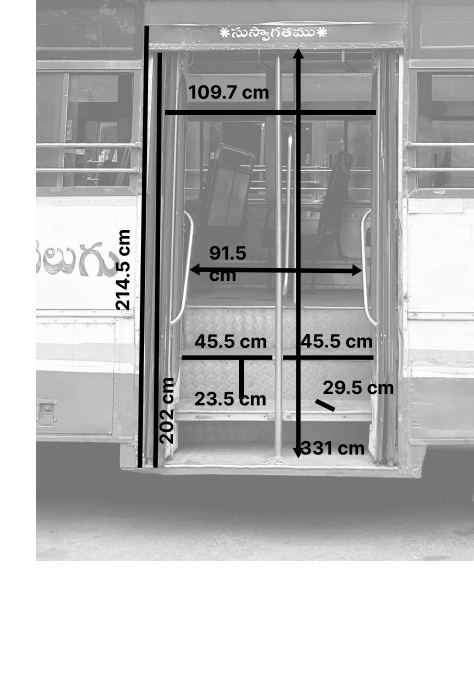
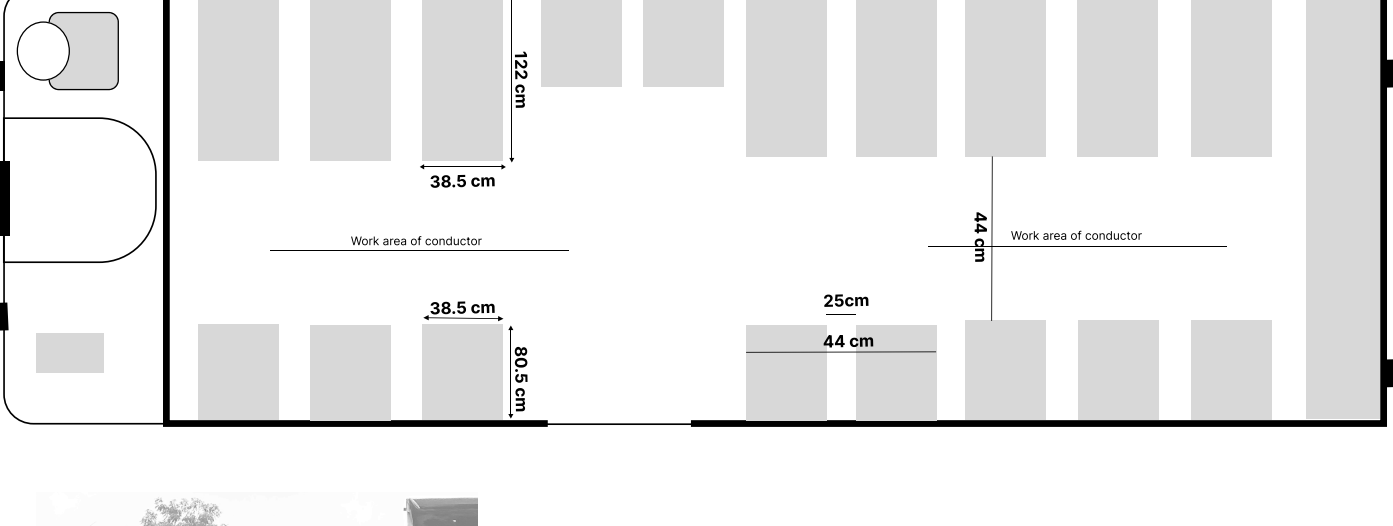
- Hip height
- Shoulder height
- Stature
- Knee height
- Shoulder-Grip length
- Thumbtip-Grip reach
- Shoulder Breadth-Biacromial
- Sitting eye height
- Sitting elbow height
- Thigh thickness/clearance
- Buttock-Popliteal length
- Hip breadth
- Elbow-Fingertip length
- Body weight
- Forward grip reach
- Elbow span

Documentation

Phone cameras were positioned within and outside the bus to document the movement of drivers and passengers performing certain tasks in stages from different planes via photography and videography. The captured visual data served as the basis for formulating the REBA (Rapid Entire Body Assessment). Additionally, two observers were stationed outside the bus to document the ingress and egress of individuals, capturing their movements for analysis.This approach facilitated a thorough understanding of postures and movements, contributing to the assessment and potential enhancement of ergonomic conditions for both drivers and passengers.

A direct engagement method, in the form of, face-to-face interviews, was employed to gather comprehensive insights from the diverse pool of Pallevelugu bus drivers. This approach was favored over a questionnaire due to its potential for yielding more accurate information, especially considering the drivers' wide array of backgrounds and varying educational levels. The interview session aimed to cover multiple aspects, including acquiring general demographic details such as age, gender, and years of professional experience. Moreover, it delved into their specific working hours and sought to understand their holiday patterns. This personalized approach was chosen to ensure a deeper understanding of the drivers' experiences and preferences.

Bus dimensions



Analysis

Driver's space:

From the acquired data of the reba scores it can be inferred Using the door to the driver's seat leaves an average REBA score of **11.75** which falls in the high risk category. Driver's door is less in width and height making it difficult to reach out to. Moreover, the driver's themselves rarely use the door meant for the driver's seat as they find it uncomfortable.

The REBA average for **reaching the side view mirror** is **11.25**. The distance between the mirror and driver('s seat) is the main concern.

The REBA analysis for **reaching the steering wheel** is **11**. Since, the diameter of the steering wheel is of certain length, reaching out to the extreme end of the steering wheel gets difficult.

REBA of **seat maximum forward** averages to **11**.

Seat minimum incline REBA averages to **11.25**. The height of the seat and placement of lever is the cause of the body movements.

Passenger's space:

Passengers frequently encounter the **challenge of navigating tight spaces as** they step up and down when **boarding or disembarking** from the bus.

This can be particularly difficult for individuals carrying luggage, pregnant women, senior citizens, and those with special mobility needs, often leading to discomfort and safety concerns.

When **passengers** are in the process of making the bus, the average REBA score stands at **11.75**, categorizing it as a **high-risk activity**.

On the other hand, while exiting, the REBA score drops to **7.9**, classifying it as a **medium-risk activity**, indicating the differences in physical strain experienced during these actions.

The **limited legroom between seats** can result in discomfort during longer journeys and may pose a challenge for taller passengers to find a comfortable seating position.

Conclusion:

In uncomfortable, the study has highlighted pressing ergonomic challenges for both bus drivers and passengers, with uncomfortable seating, reach-related issues, and accessibility problems being major concerns. However, addressing these issues necessitates a careful equilibrium between enhancing comfort and the practicality of modifying bus designs and associated infrastructure. It's important to note that certain factors, such as passenger outfits and age groups, were not considered in the assessment. The absence of female drivers for measurement also presented a limitation. Revamping buses to accommodate a broader range of passenger heights, while minimizing discomfort for the majority, poses a complex task. Altering bus dimensions would have cascading effects on city infrastructure, demanding costly and time-consuming changes. The impact on the urban landscape, public expenditure, and potential disruptions during the redesign process must be thoroughly considered, making this a formidable challenge.

A COMPREHENSIVE ANALYSIS OF SPATIAL UTILIZATION AND RISK IN A STATE TRANSPORT BUS THROUGH ACTIVITY MAPPING OF DRIVERS AND PASSENGERS .

INTRODUCTION

Research focuses on the vital role of buses in urban transit, addressing health and comfort for passengers and drivers. It investigates bus spatial design, activity analysis, and proposes interventions for improved ergonomics, highlighting health risks and providing practical recommendations.

ANTHROPO-METRIC MEASURES OF DRIVER AND PASSENGER

- Hip height (5)
- Shoulder height (3)
- Stature (1)
- Knee height (15)
- Shoulder-Grip length (25)
- Thumbtip-Grip reach (34)
- Shoulder Breadth-Biacromial (18)
- Sitting eye height (9)
- Sitting elbow height (11)
- Thigh thickness/clearance (12)
- Buttock-Popliteal length (14)
- Hip breadth (19)
- Elbow-Fingertip length (23)
- Body weight (37)
- Forward grip reach (36)
- Elbow span (33)

REBA BASED TASK ANALYSIS

TASK	Description	Avg. of REBA analysis	Risk level	Action Taken
task 1	Driver Ingress	11.8	Very High	Necessary now
task 2	Mirror adjustment	12	Very High	Necessary now
task 3	Steering extreme reach	9.6	High	Necessary soon
task 4	Seat Max. forward	8.8	High	Necessary soon
task 5	Seat Min. recline	10.8	High	Necessary soon
task 6	Passenger Ingress	13.1	Very High	Necessary now
task 7	Passenger Egress	8.4	High	Necessary soon
	AVERAGE	10.6	High	Necessary soon

METHODOLOGY WORK FLOW

SPATIAL MEASURES OF DRIVER AND PASSENGER SPACES